

2	(a) (on melting,) bonds between molecules are broken/weakened or molecules further apart/are able to slide over one another kinetic energy unchanged so no temperature change potential energy increased/changed so energy required	B1 B1 B1	[3]
	(b) thermal energy/heat required to convert unit mass of solid to liquid with no change in temperature/ at its normal boiling point	M1 A1	[2]
	(c) (i) thermal energy lost by water = $0.16 \times 4.2 \times 100$ = 67.2 kJ $67.2 = 0.205 \times L$ $L = 328 \text{ kJ kg}^{-1}$	C1 C1 A1	[3]
	(ii) more energy (than calculated) melts ice so, (calculated) L is lower than the accepted value	M1 A1	[2]

3 (a) field strength = potential gradient M1